CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 53 (2016) 90–96

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessments of the Au–Gd and Au–Yb systems

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ARTICLE INFO

Article history:
Received 22 July 2015
Received in revised form
15 February 2016
Accepted 19 February 2016
Available online 16 April 2016

Keywords:
Thermodynamic assessment
CALPHAD
Thermodynamic modeling
Phase diagram
Au–Gd binary system
Au–Yb binary system

ABSTRACT

The thermodynamic assessments of the Au–Gd and Au–Yb binary systems have been carried out by the Calculation of Phase Diagram (CALPHAD) method based on the available experimental data. The Gibbs free energies of the solution phases including liquid, fcc, bcc, and hcp were described by the substitutional solution models with the Redlich–Kister equation. All the intermetallic compounds except the βAuYb phase were treated as stoichiometric phases, and the βAuYb phase was modeled using the sublattice model. The thermodynamic parameters of each phase in the Au–Gd and Au–Yb binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.

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1. Introduction

As convenient electronic packing materials, the Sn–Pb–based solders can cause serious harm to health and environment because of the toxic nature of plumbum. Therefore, the development of Pb-free solders, which should satisfy the criterion regarding plentiful supply, favorable physical and chemical properties and cost benefit, has been considered necessarily [1,2]. As the most promising substitute for leaded solders, Sn–Au eutectic solder has been vastly employed for optoelectronic packages in high-temperature environments with high thermal and electrical conductivities, superior creep resistance and excellent solderability [3,4]. Furthermore, the addition of the appropriate amount of rare earth (RE) elements can refine the microstructure of lead-free solder alloy, which improves the wettability, mechanical property and thermal fatigue behavior of lead-free solder joint significantly [5,6]. As a result, to design the proper alloy compositions, understanding of the phase equilibria of the Sn–Au–RE becomes necessary. In the research program, the thermodynamic database for Au–RE (RE=La, Ce, Pr, Nd, Eu, Gd, Dy, Er, Ho….) will be developed.

Recently, the Au–Gd binary system was thermodynamically optimized in reference [7], where the enthalpies of mixing of liquid phase [8] and activities of components [9] in Au–Gd binary system were not considered. In the present work, the thermodynamic assessments of Au–Gd and Au–Yb systems have been carried out by the Calculation of Phase Diagram (CALPHAD) approach [10], and the thermodynamic database of the Au–RE systems have been developed, which will provide important thermodynamic information for the multicomponent Sn–Au–RE alloy systems.

2. Thermodynamic models

2.1. Liquid phases and solid solutions

The Gibbs free energies of the liquid, fcc, bcc, and hcp phases in the Au–RE (RE=Gd and Yb) binary systems are described by the substitutional solution model with using the Redlich–Kister formula [11]. The molar Gibbs free energies of solution phase in the Au–RE systems are expressed as:

$$G_m^{\Phi} = \sum_{i=Au,RE} {}^0G_i^{\Phi} x_i + RT \sum_{i=Au,RE} x_i \ln x_i + x_{Au} x_{RE} \sum_{m=0}^{n} {}^mL_{Au,RE}^{\Phi}(x_{Au} - x_{RE}). \quad (1)$$

where ${}^0G_i^{\Phi}$ is the Gibbs free energy of pure element i in the respective reference state with the Φ phase, which is taken from the SGTE pure elements database [12], R is the gas constant, T is the temperature, x_i is the molar fraction of component i. ${}^mL_{Au,RE}^{\Phi}$ is the binary interaction parameter and formulated as:

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http://dx.doi.org/10.1016/j.calphad.2016.02.002
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$$^{m}L^{\Phi}_{Au,RE} = a_m + b_m T. \quad (2)$$

where the values of a_m and b_m are optimized in the present work.

2.2. Stoichiometric compounds

All the intermetallic compounds except the βAuYb phase in the Au–RE (RE = Gd and Yb) binary systems are treated as stoichiometric phases. The Gibbs free energy per mole of formula unit Au_mRE_n can be expressed, as follows:

$$\Delta^0 G_f^{Au_mRE_n} = {}^0G_f^{Au_mRE_n} - m\,{}^0G_{Au}^{SER} - n\,{}^0G_{RE}^{SER} = a'_m + b'_m T + c'_m T \ln T + d'_m T^2 + e'_m T^3 \quad (3)$$

where the $\Delta^0 G_f^{Au_mRE_n}$ indicates the standard Gibbs free energy of formation of the stoichiometric compounds from pure elements. The parameters of a'_m, b'_m, c'_m, d'_m and e'_m are assessed in the present work.

2.3. Intermetallic compound with solubility

The intermetallic compound of the βAuYb phase with certain solubility is modeled by the two-sublattice model $(Au)_{0.5}$:$(Au, Yb)_{0.5}$. The molar Gibbs free energy of the compounds is presented by the following equation:

$$G_m^{\beta AuYb} = y^I_{Au}\,{}^0G^{\beta AuYb}_{Au:Au} + y^I_{Yb}\,{}^0G^{\beta AuYb}_{Au:Yb} + 0.5RT(y^I_{Au}\ln y^I_{Au} + y^I_{Yb}\ln y^I_{Yb}) + 0.5 y^I_{Au} y^I_{Yb} L_{Au:Au,Yb} \quad (4)$$

where y^I_i is the site fraction of element i in the first sublattice. The ${}^0G^{\beta AuYb}_{i:j}$ represents the Gibbs free energies of the βAuYb compound when the first and second sublattices are respectively occupied by elements i and j. $L_{Au:Au,Yb}$ is the interaction energy between Au atoms in the first sublattice when the second sublattice is occupied by Au and Yb, and is expressed as:

$$L^{\varphi}_{Au:Au,Yb} = a'' + b''T + c''\ln T \quad (5)$$

the parameters of a'', b'' and c'' are evaluated in this work.

3. Thermodynamic optimization

The optimization of the thermodynamic parameters was carried out by using PARROT program in the Thermo-Calc software [13], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the available experimental data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

3.1. The Au–Gd system

3.1.1. Experimental information

In the Au–Gd system, the terminal solid solubility of Gd in (Au) was experimentally determined to be 0.7 at% Gd by Rider et al. [14], and the intermetallic compounds and the melting point of βAuGd phase was presented by McMasters et al. [15]. Besides these, the crystal structures of the intermetallic compounds (Au_6Gd [15,16], $Au_{51}Gd_{14}$ [15], Au_3Gd [17], Au_2Gd [15,18,19], AuGd (β and α allotropic forms) [15,20,21] and $AuGd_2$ [15]) were obtained. Later, the phase diagram in the Au–Gd system was studied by Saccone et al. [22], who firstly reported the compounds of $Au_{10}Gd_7$ and Au_4Gd_3 phases, and identified four solution phases (liquid, (Au), (αGd), (βGd)) and 9 intermetallic compounds (Au_6Gd, $Au_{51}Gd_{14}$, Au_3Gd, Au_2Gd, $Au_{10}Gd_7$, Au_4Gd_3,

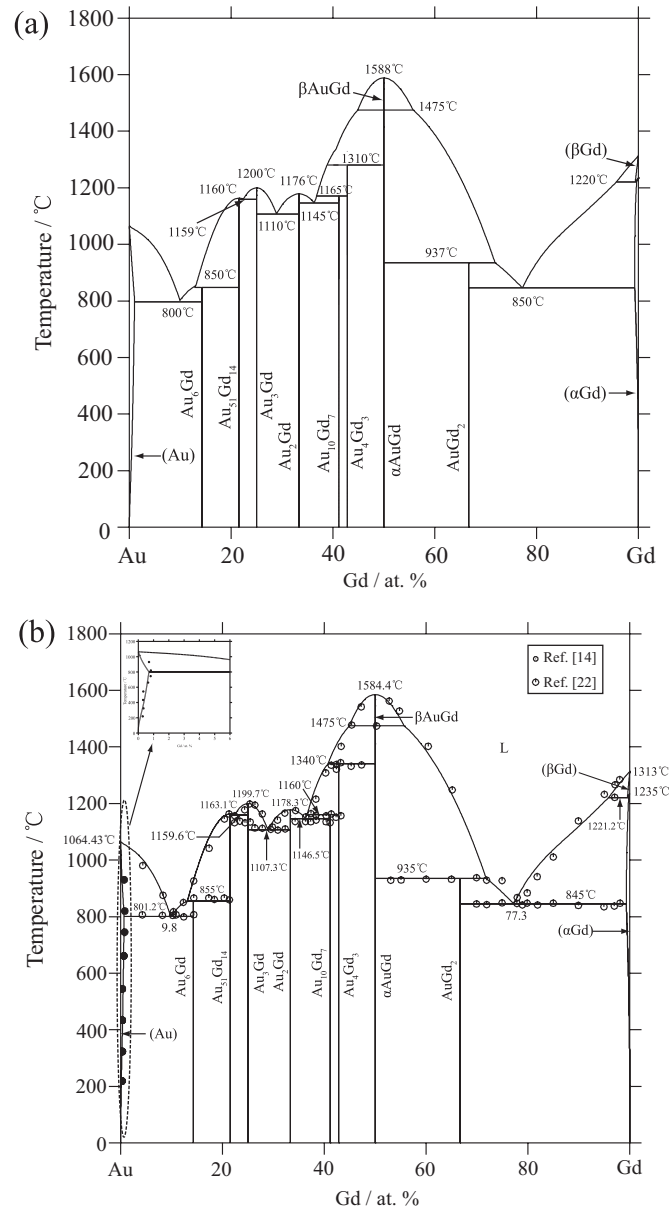

Fig. 1. Calculated phase diagram of the Au–Gd system: (a) from Ref. [7]; (b) in this work compared with the experimental data [14,22].

βAuGd, αAuGd and $AuGd_2$) with differential thermal analysis (DTA), X-ray diffraction (XRD), optical and scanning electron microscopy (SEM), and electron probe microanalysis (EPMA). Saccone et al. [22] also determined five eutectic reactions (L↔(Au)+Au_6Gd at 11.0 at% Gd and 800 °C, L↔$Au_{51}Gd_{14}$+Au_3Gd at 23.0 at% Gd and 1135 °C, L↔Au_3Gd+Au_2Gd at 29.5 at% Gd and 1110 °C, L↔Au_2Gd+$Au_{10}Gd_7$ at 36.5 at% Gd and 1140 °C, and L↔$AuGd_2$+(αGd) at 79.0 at% Gd and 845 °C), four peritectic reactions (L+$Au_{51}Gd_{14}$↔Au_6Gd at 855 °C, L+Au_4Gd_3↔$Au_{10}Gd_7$ at 1160 °C, L+αAuGd↔Au_4Gd_3 at 1340 °C, and L+αAuGd↔$AuGd_2$ at 935 °C), a polymorphic reaction (βAuGd↔αAuGd at 1475 °C) and a catatectic reaction ((βGd)↔L+(αGd) at 1220 °C).

Additionally, Fitzner and Kleppa [23] experimentally determined the value of standard enthalpies of formation of αAuGd, Au_2Gd and Au_3Gd compounds by using high-temperature direct synthesis calorimety is −82.6 ± 2.7 kJ/mol of atoms, −77.2 ± 2.7 kJ/mol of atoms, −69.9 ± 2.2 kJ/mol of atoms respectively. Later, Ferro et al. [24] reported the value of standard enthalpies of formation of αAuGd phase is −82.0 ± 2.0 kJ/mol of atoms at 27 °C. Witusiewicz and Ivanov [8]

Table 1
Comparison between the calculated and experimental results [14,22] in the Au–Gd system.

Invariant Reaction	Reaction type	Composition (at% Gd)			Temperature (°C)	References
$(\beta Gd) \leftrightarrow (\alpha Gd) + L$	Catatectic	97.0	≥ 99.0	95.0	1220	[22]
		99.6	99.8	95.8	1221.2	This work
$L \leftrightarrow (\alpha Gd) + AuGd_2$	Eutectic	79.0	≥ 99.0	66.7	845	[22]
		77.3	99.3	66.7	845	This work
$L + \alpha AuGd \leftrightarrow AuGd_2$	Peritectic	74.0	50.0	66.7	935	[22]
		71.8	50.0	66.7	935	This work
$L \leftrightarrow \beta AuGd$	Congruent	–	50.0	–	1585	[22]
		–	50.0	–	1584.4	This work
$\beta AuGd \leftrightarrow \alpha AuGd$	Allotropic	–	50.0	–	1475	[22]
		–	50.0	–	1475	This work
$L + \alpha AuGd \leftrightarrow Au_4Gd_3$	Peritectic	41.0	50.0	42.9	1340	[22]
		40.7	50.0	42.9	1340	This work
$L + Au_4Gd_3 \leftrightarrow Au_{10}Gd_7$	Peritectic	37.0	42.9	41.2	1160	[22]
		36.6	42.9	41.2	1160	This work
$L \leftrightarrow Au_{10}Gd_7 + Au_2Gd$	Eutectic	36.5	41.2	33.3	1140	[22]
		36.3	41.2	33.3	1146.5	This work
$L \leftrightarrow Au_2Gd$	Congruent	–	33.3	–	1180	[22]
		–	33.3	–	1178.3	This work
$L \leftrightarrow Au_2Gd + Au_3Gd$	Eutectic	29.5	33.3	25.0	1110	[22]
		28.9	33.3	25.0	1107.3	This work
$L \leftrightarrow Au_3Gd$	Congruent	–	25.0	–	1200	[22]
		–	25.0	–	1199.7	This work
$L \leftrightarrow Au_3Gd + Au_{51}Gd_{14}$	Eutectic	23.0	25.0	21.5	1135	[22]
		22.4	25.0	21.5	1159.6	This work
$L \leftrightarrow Au_{51}Gd_{14}$	Congruent	–	21.5	–	1165	[22]
		–	21.5	–	1163.1	This work
$L + Au_{51}Gd_{14} \leftrightarrow Au_6Gd$	Peritectic	12.0	21.5	14.3	855	[22]
		13.1	21.5	14.3	855	This work
$L \leftrightarrow Au_6Gd + (Au)$	Eutectic	–	–	0.7	809	[14]
		11.0	14.3	0.7	800	[22]
		9.8	14.3	0.7	801.2	This work

performed high temperature calorimetric method in the liquid binary alloy at 1357 ± 10 °C, and reported the partial mixing enthalpies of liquid phase in the Au–Gd binary system. With the Knudsen effusion method, Ivanov and Berezutski [9] measured the thermodynamic activity of Au in the liquid Au–Gd alloys at 1350 °C over a wide composition range except for the heterogeneous field around the AuGd compound. And the activity of Gd was calculated by the Gibbs–Duhem equation [9].

3.1.2. Optimized results and discussion

As shown in Fig. 1, compared with the previous work [7], the reassessed phase diagram in the present work is more reasonably in agreement with the experimental data. Table 1 presents the compositions of the phases and the temperatures of all the invariant reactions. It is seen that the good agreement is obtained between the calculated results and most experimental information except for the temperature of the eutectic reaction ($L \leftrightarrow Au_3Gd + Au_{51}Gd_{14}$), where the discrepancy between the present evaluation and the proposed experimental data [22] is 24.6 °C. Okamoto and Massalski [25] indicated that the width on one side is either one half or double that on the other side, the asymmetry may be considered clearly unusual. With regard to Au_3Gd phase, the ratio is significantly more two times in the experimental phase diagram, thus it should be modified. Accordingly, the calculated temperature of this eutectic reaction has to be sacrificed in order to adjust to the liquidus symmetry and the enthalpy of formation of the Au_3Gd compound. On the other hand, Okamoto and Massalski [25] also mentioned that a compound with sharp liquidus will decompose eutectoidally at low temperature. Since no experimental information indicates that Au_3Gd phase decomposes at low temperature, the reported liquidus should be flatter. Therefore, the discrepancy between the calculated temperature of the eutectic reaction ($L \leftrightarrow Au_3Gd + Au_{51}Gd_{14}$) and experimental data [22] can be acceptable. Table 2 lists a complete set of self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Au–Gd system.

Considering the thermochemical information, the standard enthalpies of formation of the intermetallic compounds in the Au–Gd system are reproduced by the thermodynamical parameters. Fig. 2 illustrates a reasonable agreement between the optimized results and the experimental data [23,24]. The calculated mixing enthalpies of liquid phase as depicted in Fig. 3 are accordant to the experimental data [8] basically.

The calculated activities of Au and Gd compared with the experimental data [9] in the liquid Au–Gd binary alloys at 1350 °C with reference states of pure elements of Au (liquid) and Gd (liquid) are illustrated in Fig. 4(a) and (b). It can be seen that the calculated activity of Au is in good agreement with the experimental data [9] in the Au-rich side, while a large divergence is noted with the experimental data [9] in the Gd-rich side. Due to the high reactivity of Gd, it is difficult to accurately measure the activity of the alloy with more than 50 at% Gd by the Knudsen effusion technique. Hence, the difference between the calculated activity of Au and the experimental data [9] in Gd-rich side is considered reasonable to some extent. And the calculated activity of Gd fits with the calculated data [9] basically in Fig. 4(a) and (b).

3.2. The Au–Yb system

3.2.1. Experimental information

In the Au–Yb system, the terminal solid solubility of Yb in (Au) was reported by Rider et al. [14] with using X-ray parametric method, and the compounds of AuYb [19], Au_2Yb [26], Au_3Yb [27] and Au_4Yb [28] were identified. Later, with differential thermal, micrographic and X-ray methods, Iandelli and Palenzona [28] investigated the phase diagram of Au–Yb system and reported eight intermediate compounds (Au_4Yb, Au_3Yb, Au_2Yb, AuYb (β and α allotropic forms), Au_4Yb_5, Au_3Yb_5, $AuYb_2$ and Au_3Yb_7), four

Table 2
Thermodynamic parameters in the Au–Gd systems assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Au, Gd)
${}^{0}L^{Liq}_{Au,Gd} = -293000 + 9.5T$
${}^{1}L^{Liq}_{Au,Gd} = -112500 - 9.5T$
${}^{2}L^{Liq}_{Au,Gd} = 47300 + 1.5T$
${}^{3}L^{Liq}_{Au,Gd} = 71600 - 6.2T$
Fcc phase, format (Au, Gd)
${}^{1}L^{Fcc}_{Au,Gd} = -253285$
Bcc phase, format (Au, Gd)
${}^{0}L^{Bcc}_{Au,Gd} = -137515$
Hcp phase, format (Au, Gd)
${}^{0}L^{hcp}_{Au,Gd} = -127300$
Au_6Gd phase, format $(Au)_{0.8571}(Gd)_{0.1429}$
$Au_{51}Gd_{14}$ phase, format $(Au)_{0.7846}(Gd)_{0.2154}$
${}^{0}G^{Au_{51}Gd_{14}}_{Au:Gd} - 0.7846{}^{0}G^{fcc}_{Au} - 0.2154{}^{0}G^{hcp}_{Gd} = -57150 + 1.75T - 0.648T\mathrm{Ln}(T)$
Au_3Gd phase, format $(Au)_{0.75}(Gd)_{0.25}$
${}^{0}G^{Au_3Gd}_{Au:Gd} - 0.75{}^{0}G^{fcc}_{Au} - 0.25{}^{0}G^{hcp}_{Gd} = -63500 + 9.65T - 1.745T\mathrm{Ln}(T)$
Au_2Gd phase, format $(Au)_{0.6667}(Gd)_{0.3333}$
$Au_{10}Gd_7$ phase, format $(Au)_{0.5882}(Gd)_{0.4118}$
Au_4Gd_3 phase, format $(Au)_{0.5714}(Gd)_{0.4286}$
${}^{0}G^{Au_4Gd_3}_{Au:Gd} - 0.5714{}^{0}G^{fcc}_{Au} - 0.4286{}^{0}G^{hcp}_{Gd} = -76940 - 3.878T$
βAuGd phase, format $(Au)_{0.5}(Gd)_{0.5}$
${}^{0}G^{AuGd}_{Au:Gd} - 0.5{}^{0}G^{fcc}_{Au} - 0.5{}^{0}G^{hcp}_{Gd} = -77550 - 3.0366T$
αAuGd phase, format $(Au)_{0.5}(Gd)_{0.5}$
$AuGd_2$ phase, format $(Au)_{0.3333}(Gd)_{0.6667}$
${}^{0}G^{AuGd_2}_{Au:Gd} - 0.3333{}^{0}G^{fcc}_{Au} - 0.6667{}^{0}G^{hcp}_{Gd} = -53208 - 3.1357T$

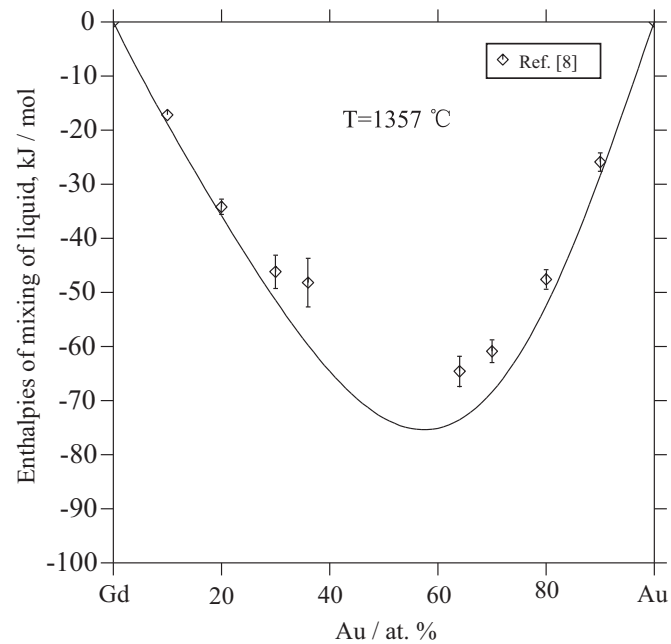

Fig. 3. Calculated enthalpies of mixing of liquid phase at 1357 °C with reference states of pure elements of Au (liquid) and Gd (liquid) compared with the experimental data [8].

eutectic reactions (L↔(βYb)+Au_3Yb_7 at 81.5 at% Yb and 546 °C, L↔βAuYb+Au_2Yb at 41.5 at% Yb and 1061 °C, L↔ Au_2Yb+ Au_3Yb at 28.0 at% Yb and 1131 °C, and L↔ Au_4Yb+(Au) at 11.0 at% Yb and 809 °C), five peritectic reactions (L+$AuYb_2$↔Au_3Yb_7 at 661 °C, L+Au_3Yb_5↔$AuYb_2$ at 712 °C, L+Au_4Yb_5↔Au_3Yb_5 at 861 °C, L+βAuYb↔Au_4Yb_5 at 926 °C, and L+Au_3Yb↔Au_4Yb at 989 °C), and a polymorphic reaction (βAuYb↔αAuYb at 470 °C).

In addition, Ferro et al. [24] experimentally measured the value of the standard enthalpies of formation of αAuYb phase is −75.0 ± 2.0 kJ/mol of atoms by the high-temperature direct

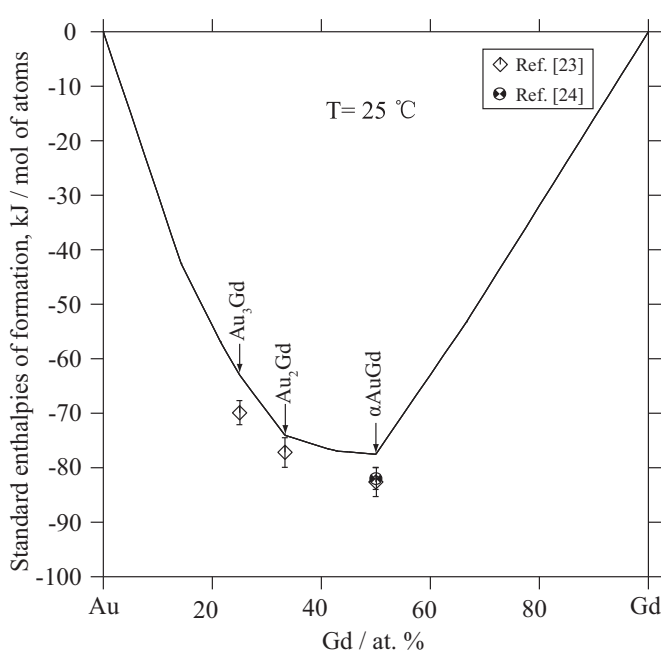

Fig. 2. Calculated standard enthalpies of formation with reference states of Au (fcc) and αGd (hcp) compared with the experimental data [23,24].

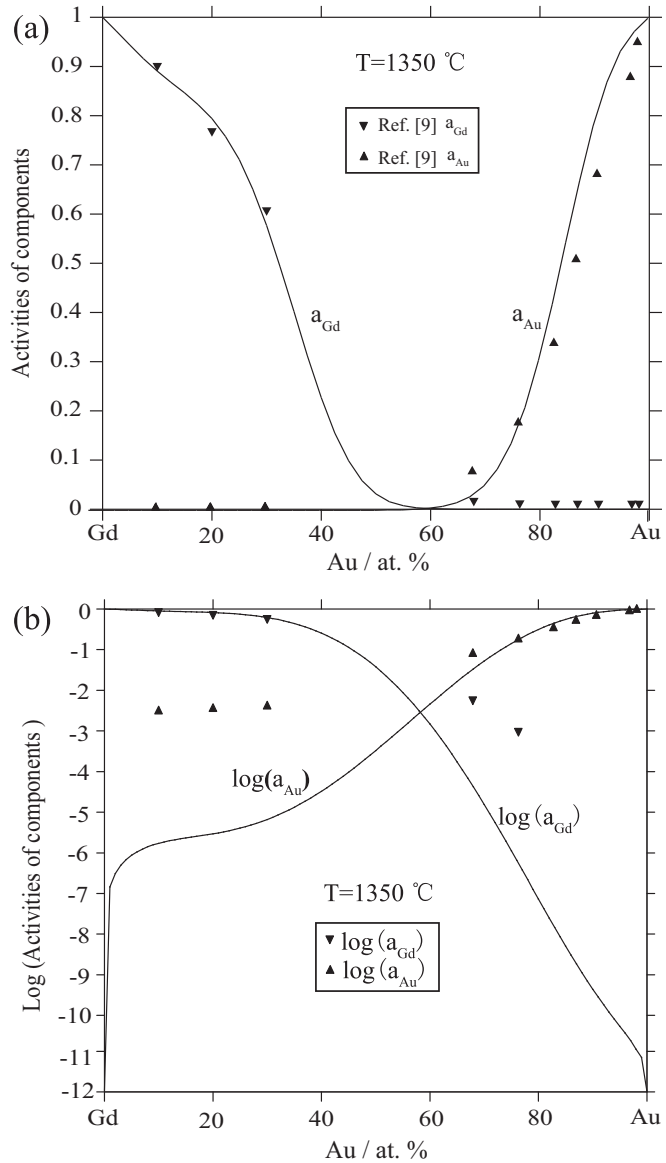

Fig. 4. Calculated activities of components at 1350 °C in liquid Au–Gd system with reference states of pure elements of Au (liquid) and Gd (liquid) compared with the experimental data [9]. (a) linear scale, (b)log scale.

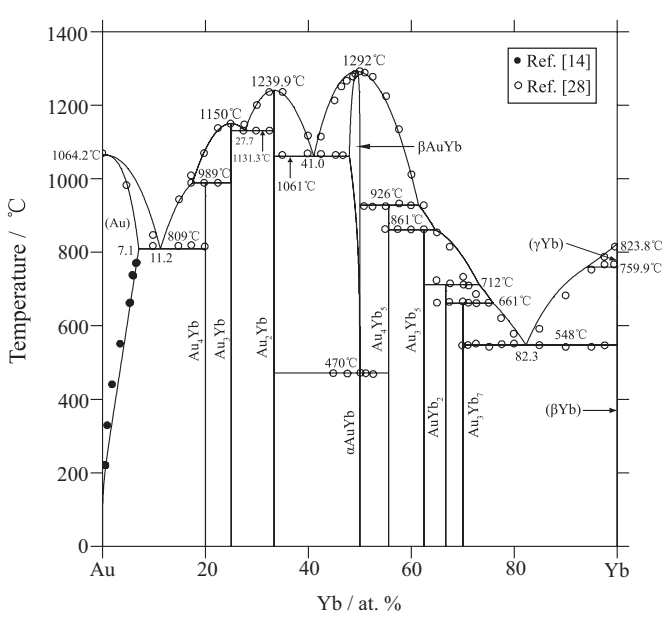

Fig. 5. Calculated phase diagram of the Au–Yb system in this work compared with the experimental data [14,28].

Table 3
Comparison between the calculated and experimental results [14,28] in the Au–Yb system.

Invariant reaction	Reaction type	Composition (at% Yb)			Temperature (°C)	Reference
$L \leftrightarrow (\beta Yb) + Au_3Yb_7$	Eutectic	81.5	–	70.0	546	[28]
		82.3	100	70.0	548	This work
$L + AuYb_2 \leftrightarrow Au_3Yb_7$	Peritectic	–	66.7	70.0	661	[28]
		75.9	66.7	70.0	661	This work
$L + Au_3Yb_5 \leftrightarrow AuYb_2$	Peritectic	–	62.5	66.7	712	[28]
		73.1	62.5	66.7	712	This work
$L + Au_4Yb_5 \leftrightarrow Au_3Yb_5$	Peritectic	–	55.6	62.5	861	[28]
		64.6	55.6	62.5	861	This work
$L + \beta AuYb \leftrightarrow Au_4Yb_5$	Peritectic	–	50.0	55.6	926	[28]
		61.4	50.0	55.6	926	This work
$L \leftrightarrow \beta AuYb$	Congruent	–	50.0	–	1292	[28]
		–	50.0	–	1292	This work
$\beta AuYb \leftrightarrow \alpha AuYb$	Allotropic	–	50.0	–	470	[28]
		–	50.0	–	470	This work
$L \leftrightarrow \beta AuYb + Au_2Yb$	Eutectic	41.5	50.0	33.3	1061	[28]
		41.0	50.0	33.3	1061	This work
$L \leftrightarrow Au_2Yb$	Congruent	–	33.3	–	1240	[28]
		–	33.3	–	1239.9	This work
$L \leftrightarrow Au_2Yb + Au_3Yb$	Eutectic	28.0	33.3	25.0	1131	[28]
		27.7	33.3	25.0	1131.3	This work
$L \leftrightarrow Au_3Yb$	Congruent	–	25.0	–	1150	[28]
		–	25.0	–	1150	This work
$L + Au_3Yb \leftrightarrow Au_4Yb$	Peritectic	–	25.0	20.0	989	[28]
		17.5	25.0	20.0	989	This work
$L \leftrightarrow Au_4Yb + (Au)$	Eutectic	–	–	6.9	820	[14]
		11.0	20.0	6.9	809	[28]
		11.2	20.0	7.1	809	This work

Table 4
Thermodynamic parameters in the Au–Yb system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Au, Yb)
${}^0L^{Liq}_{Au,Yb} = -226000 - 32T + 7.79T\mathrm{Ln}(T)$
${}^1L^{Liq}_{Au,Yb} = -53500 - 3T$
${}^2L^{Liq}_{Au,Yb} = 35300 - 25T$
${}^3L^{Liq}_{Au,Yb} = -13050 + 7T$
Fcc phase, format (Au, Yb)
${}^0L^{Fcc}_{Au,Yb} = -137036 + 13T$
${}^1L^{Fcc}_{Au,Yb} = -68720 - 35T$
${}^2L^{Fcc}_{Au,Yb} = 25950 - 36T$
Au_4Yb phase, format $(Au)_{0.8}(Yb)_{0.2}$
${}^0G^{Au_4Yb}_{Au:Yb} - 0.8{}^0G^{fcc}_{Au} - 0.2{}^0G^{hcp}_{Yb} = -46800 + 3.7679T$
Au_3Yb phase, format $(Au)_{0.75}(Yb)_{0.25}$
${}^0G^{Au_3Yb}_{Au:Yb} - 0.75{}^0G^{fcc}_{Au} - 0.25{}^0G^{hcp}_{Yb} = -52376 + 2.1077T$
Au_2Yb phase, format $(Au)_{0.6667}(Yb)_{0.3333}$
${}^0G^{Au_2Yb}_{Au:Yb} - 0.6667{}^0G^{fcc}_{Au} - 0.3333{}^0G^{hcp}_{Yb} = -56816 + 0.1098T$
$\beta AuYb$ phase, format $(Au)_{0.5}(Au, Yb)_{0.5}$
${}^0G^{AuYb}_{Au:Au} - {}^0G^{fcc}_{Au} = 5000$
${}^0G^{AuYb}_{Au:Yb} - 0.5{}^0G^{fcc}_{Au} - 0.5{}^0G^{hcp}_{Yb} = -59000 + 0.0623T$
${}^0G^{AuYb}_{Au:Au,\,Yb} = -16310 + 31T - 7T\mathrm{Ln}(T)$
$\alpha AuYb$ phase, format $(Au)_{0.5}(Yb)_{0.5}$
${}^0G^{AuYb}_{Au:Yb} - 0.5{}^0G^{fcc}_{Au} - 0.5{}^0G^{hcp}_{Yb} = -70000 + 14.8641T$
Au_4Yb_5 phase, format $(Au)_{0.4444}(Yb)_{0.5556}$
${}^0G^{Au_4Yb_5}_{Au:Yb} - 0.4444{}^0G^{fcc}_{Au} - 0.5556{}^0G^{hcp}_{Yb} = -62010 + 5.9632T$
Au_3Yb_5 phase, format $(Au)_{0.375}(Yb)_{0.625}$
${}^0G^{Au_3Yb_5}_{Au:Yb} - 0.375{}^0G^{fcc}_{Au} - 0.625{}^0G^{hcp}_{Yb} = -54000 + 4.1101T$
$AuYb_2$ phase, format $(Au)_{0.333}(Yb)_{0.667}$
${}^0G^{AuYb_2}_{Au:Yb} - 0.3333{}^0G^{fcc}_{Au} - 0.6667{}^0G^{hcp}_{Yb} = -49200 + 4.0906T$
Au_3Yb_7 phase, format $(Au)_{0.3}(Yb)_{0.7}$
${}^0G^{Au_3Yb_7}_{Au:Yb} - 0.3{}^0G^{fcc}_{Au} - 0.7{}^0G^{hcp}_{Yb} = -44650 + 3.6044T$

synthesis calorimetry. The mixing enthalpies of liquid phase in the Au–Yb binary system at 1190 °C was determined in the range of 0–45 at% Yb by Ivanov et al. [29]. Moreover, de Boer and Pettifor [30] predicted the value of the enthalpies of mixing of liquid phase at 50 at% Yb in the Au–Yb binary system is −59 kJ/mol.

3.2.2. Optimized results and discussion

The calculated Au–Yb phase diagram with the experimental data [14,28] is shown in Fig. 5. Table 3 presents the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data [14,28]. It is seen that the calculated results well reproduced the experimental data [14,28]. A complete set of self-consistent thermodynamic

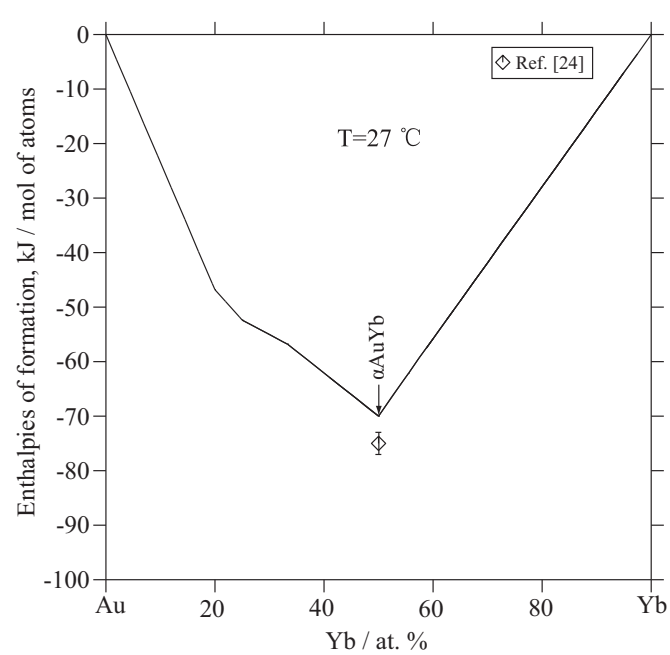

Fig. 6. Calculated standard enthalpies of formation with reference states of Au (fcc) and βYb (fcc) compared with the experimental data [24].

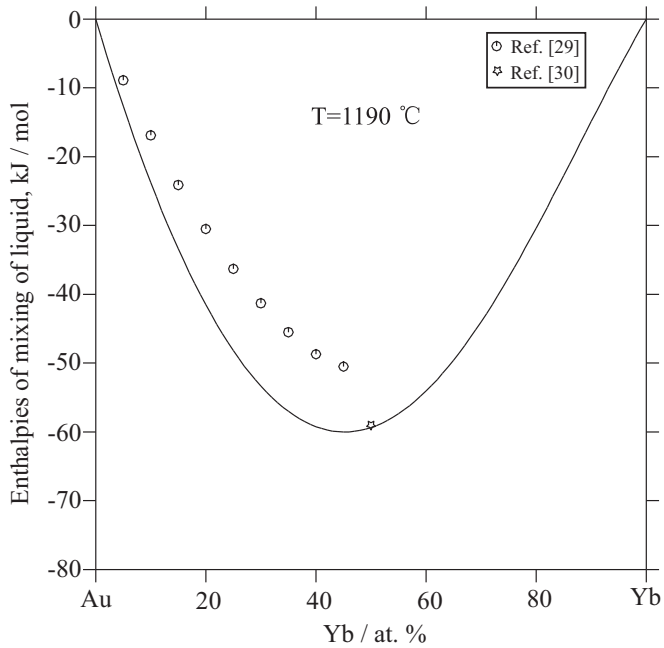

Fig. 7. Calculated enthalpies of mixing of liquid phase at 1190 °C with reference states of pure elements of Au (liquid) and Yb (liquid) compared with the experimental data [29] and predicted data [30].

parameters describing the Gibbs free energy of each phase in the Au–Yb system are given in Table 4.

As shown in Fig. 6, the calculated standard enthalpies of formation with reference states of fcc (Au) and fcc (βYb) phases compared with the experimental data [24] are observed and considered acceptable. The calculated mixing enthalpies of liquid phase at 1190 °C with reference states of pure elements of Au (liquid) and Yb (liquid) in the Au–Yb system is illustrated in Fig. 7. It is seen that the calculated results are slightly more negative in the concentration range of 0–45 at% Yb than the experimental data [29]. However, the calculated value of the mixing enthalpy of liquid phase at 50 at% Yb is −56.6 kJ/mol, which is close to the predicated data [30] −59 kJ/mol. As described in Fig. 7, the enthalpies of mixing of liquid phase assessed in this work show the general tendency similar to experimental data [29].

4. Conclusions

On the basis of available experimental data, the Au–RE (RE = Gd and Yb) binary systems have been critically optimized by the CALPHAD method. The thermodynamic parameters of each phase in the Au–Gd and Au–Yb binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.

Acknowledgments

This work was supported by the Open Fund (PLN1306)of State Key Laboratory of Oil and Gas Reservoir Geology and Exploitation (Southwest Petroleum University), National Natural Science Foundation of China (51374180), Youth Science and Technology Innovation Team of Electrochemistry for Energy Materials, Southwest Petroleum University (2015CXTD04), Scientific Research Foundation and Opening Foundation (X151515KCL04 and X151515KCL06) of Southwest Petroleum University, National Undergraduate Training Program for Innovation and Entrepreneurship (201510615034), and Educational Commission (15ZA0052) of Sichuan Province of China.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2016.02.002.

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